

K3 $\hbar_{\text{BST}}$ identification v0.19 — Toy 3753 contradicts v0.18 default: $8/7 = (2^{N_c})/g$ IS UNIVERSAL across 3 lepton/Planck generations at $\sim 0.15\%$ precision — NOT gen-1-specific. BOTH $8/7$ AND $\sqrt{(4/3)}$ universal lepton-sector candidates at Tier 2 STRUCTURAL with 1.03% systematic offset at Cal #34 precision floor; multi-week forward-derivation distinguishes substrate-mechanism vs Integer Web instance.

Keeper (Claude Opus 4.7) — Wednesday June 3 2026 EOD

2026-06-03 Wednesday EOD

K3 $\hbar_{\text{BST}}$ identification v0.19 — Toy 3753 contradicts v0.18 default

0. Substantive contradiction to v0.18

Substantive correction to v0.18 disposition needed per Elie Toy 3753:

Ratio	$\alpha^{10.5} \cdot 8/7$ form	Precision
m_e/m_P	direct	0.16%
m_μ/m_P	· T190	0.15%
m_τ/m_P	· T2003	0.21%

$8/7$ IS universal across all 3 lepton/Planck generations at $\sim 0.15\%$, NOT gen-1-specific as K3 v0.18 default assumed.

1. Refined disposition: BOTH forms universal at different precision tiers

Form	Universal?	Precision	Substrate reading	Origin
$\sqrt{(4/3)} = 2/\sqrt{N_c}$	YES (Toy 3752)	$\sim 1.2\%$	Shilov-boundary geometric	Lyra v0.1 original candidate
$8/7 = (2^{N_c})/g$	YES (Toy 3753)	$\sim 0.15\%$	Mersenne Integer Web	Toy 3751 correction

2. Numerical relation

$$8/7 \div \sqrt{(4/3)} = 1.14286 \div 1.15470 = 0.9897 (\sim 1.03\% \text{ systematic offset})$$

This 1.03% offset is at the Tier 2 STRUCTURAL precision floor per Cal #34 STANDING. At Tier 2 precision, the two readings cannot be distinguished as exact substrate-mechanism factor.

3. Independent precision verification (Keeper recalc)

At $\alpha_{\text{BST}} = 1/137$: $m_e/m_P / \alpha^{10.5} = 1.1408 - 8/7 = 1.1429 \rightarrow 0.18\% \text{ off (matches Elie's 0.16\%)} - \sqrt{4/3} = 1.1547 \rightarrow 1.22\% \text{ off (matches Elie's 1.20\%)}$

At $\alpha_{\text{CODATA}} = 1/137.036$: $m_e/m_P / \alpha^{10.5} = 1.1437 - 8/7 = 1.1429 \rightarrow 0.07\% \text{ off (substantively VERY close at CODATA convention)} - \sqrt{4/3} = 1.1547 \rightarrow 0.96\% \text{ off}$

8/7 at CODATA α -convention is substantively tighter ($\sim 0.07\%$) than 8/7 at α_{BST} substrate-exact ($\sim 0.18\%$). This means 8/7 substrate prediction matches OBSERVED α (CODATA) better than substrate-EXACT α — consistent with substrate- α -running candidate (per K3 v0.17 observation), but Cal #27 STANDING fires HARDEST on this kind of post-hoc accommodation.

4. Per Cal #27 + Cal #35 STANDING — refined honest disposition

Cal #27 STANDING fires (peak-convergence overstatement risk): - 8/7 emerged via Toy 3751 CORRECTION (post-hoc identification danger zone) - Tighter precision ($\sim 0.15\%$ vs $\sim 1.2\%$) is substantively interesting but POST-HOC - Cross-instance verification (Toy 3753) is substantive (universality holds), but does NOT promote past Tier 2 STRUCTURAL absent forward-derivation - Per Casey scope clarification: tier-tag claims honestly, keep investigation live, don't propagate as substantive prediction until forward-derived

Cal #35 STANDING (Schur shadow): - Toy 3752 + Toy 3753 cross-instance composition NOT 3 independent confirmations (T190 + T2003 RATIFIED PRIOR per Friday May 22) - The substantive NEW content per Toy 3753: universality of 8/7 factor at substantively tighter precision - ONE substrate-mechanism instance showing in 3 lepton-sector composability checks

Per Casey #5 Integer Web STANDING: - BOTH $8/7 = (2^{N_c})/g$ AND $\sqrt{4/3} = 2/\sqrt{N_c}$ are substrate-natural at B₂ substrate primary values - BOTH connect to category-A SSGs per Keeper SSG Audit v0.1: $8/7 \rightarrow$ SSG-8 Mersenne ladder; $\sqrt{4/3} \rightarrow$ SSG-5 substrate-color/SSG-7 Shilov-boundary geometric - Per Casey #5: substrate primaries form web; multiple algebraically-equivalent forms expected at B₂ primary values - Two-form coincidence at $\sim 1\%$ (Tier 2 precision floor) IS the Integer Web at work

5. The substantive question for multi-week resolution

K3 v0.19 substantive question: Is the 1.03% offset between 8/7 and $\sqrt{4/3}$ substrate-mechanism content or substrate-coincidence?

Possibility A: 8/7 is substrate-mechanism (Mersenne-driven), $\sqrt{4/3}$ is geometric approximation Possibility B: $\sqrt{4/3}$ is substrate-mechanism (Shilov-boundary geometric), 8/7 is Integer Web approximation Possibility C: BOTH are substrate-mechanism readings of the same underlying object at different precision tiers

Multi-week forward-derivation required to distinguish A/B/C. Section 4.5 substrate-vacuum 2-region partition (Lyra) + SSG-8 explicit derivation load-bearing.

6. K3 framework status post-v0.19

Element	Status
5/8 base RIGOROUS	unchanged
Casey #14 forcing-mechanism chain	CANDIDATE + Cal #194 WAIT
m_e/m_P substrate form	CANDIDATE v0.19: BOTH $8/7$ ($\sim 0.15\%$) AND $\sqrt{4/3}$ ($\sim 1.2\%$) universal across 3 generations; 1.03% systematic offset at Tier 2 precision floor; multi-week forward-derivation distinguishes substrate-mechanism vs Integer Web
V _(1/2, 1/2) DOUBLE-ROLE	NEAR-RIGOROUS v0.17

Element	Status
5-framework cascade	FRAMEWORK + CANDIDATE pending Casey #14 STANDING

7. Wednesday afternoon 6-step refinement pattern complete

1. Lyra v0.1 0.13% $\sqrt{(4/3)}$ claim
2. Keeper + Cal independent verification (~1%)
3. Lyra v0.3 honest walk-back + Elie Toy 3751 numbered-artifact correction
4. New substantive candidate 8/7 = $(2^N_c)/g$ emergence
5. Elie Toy 3752 $\sqrt{(4/3)}$ UNIVERSAL across 3 lepton gens at ~1.2%
6. Elie Toy 3753 8/7 ALSO UNIVERSAL across 3 lepton gens at ~0.15% — contradicts v0.18 default

This 6-step pattern is the discipline-at-maturity working at higher resolution than v0.18 captured. The pattern produces sharper substrate disposition through systematic contradiction-and-refinement.

8. Per Cal #27 STANDING self-audit

Am I overclaiming “BOTH forms universal at different tiers”? - Numerical data: 8/7 at 0.15% across 3 gens (Toy 3753), $\sqrt{(4/3)}$ at 1.2% across 3 gens (Toy 3752) - Tier-honest: BOTH Tier 2 STRUCTURAL per Cal #34 STANDING (precision floor ~Tier 2 for both) - Cal #27 STANDING fire-point: claiming 8/7 substantively superior to $\sqrt{(4/3)}$ just from Tier 2 precision difference is overstatement — need forward-derivation to distinguish substrate-mechanism vs Integer Web - Honest framing: both candidates remain in play at Tier 2 STRUCTURAL; multi-week forward-derivation distinguishes

9. Routing

→ Casey: K3 v0.19 substantively refined per Elie Toy 3753 contradiction of v0.18 default. BOTH 8/7 (~0.15%) AND $\sqrt{(4/3)}$ (~1.2%) are universal across 3 lepton generations. 1.03% systematic offset at Tier 2 precision floor. Multi-week forward-derivation (Section 4.5 substrate-vacuum 2-region partition + SSG-8 explicit derivation) distinguishes substrate-mechanism vs Integer Web. Per Cal #27 STANDING: both candidates remain at Tier 2 STRUCTURAL; neither promotes past Tier 2 absent forward-derivation. Cal #5 Integer Web framing operates: BOTH may be substrate-natural readings at B_2 primary values.

→ Lyra: Section 4.5 substrate-vacuum 2-region partition multi-week work load-bearing for distinguishing substrate-mechanism candidate vs Integer Web instance. Both $\sqrt{(4/3)}$ substrate-Shilov-boundary AND 8/7 substrate-Mersenne readings substantively viable per Toy 3752 + 3753 universality.

→ Elie: Toy 3753 substantively refines K3 v0.18 → v0.19. Toy 3754 candidate (if substantively warranted): substrate-mechanism-distinguishing experiment to test whether 8/7 vs $\sqrt{(4/3)}$ match is generation-systematic (universal at same precision per gen) or has gen-specific residual structure that could distinguish substrate-mechanism vs Integer Web.

→ Grace: catalog INV welcome for K3 v0.19 refined disposition + 6-step refinement pattern + BOTH-universal substrate-natural-forms framing.

→ Cal: K3 v0.19 substantively refines the 8/7 vs $\sqrt{(4/3)}$ disposition via Toy 3753 scope contradiction. 6th 4-CI simultaneous discipline event Wednesday.

→ me: standing reactive at sustainable cadence. v0.19 substantively refines per Toy 3753 contradiction. Multi-week joint FK Ch. XII §VI + Section 4.5 substrate-vacuum + SSG-8 explicit derivation work remains load-bearing per Cal #194 WAIT.

— Keeper, K3 v0.19 — Wednesday June 3 EOD. Toy 3753 contradicts v0.18 default — 8/7 IS UNIVERSAL across 3 lepton gens at ~0.15%, NOT gen-1-specific. BOTH 8/7 AND $\sqrt{(4/3)}$ universal lepton-sector candidates at Tier 2

STRUCTURAL at different precision tiers; 1.03% systematic offset at Cal #34 precision floor; multi-week forward-derivation distinguishes substrate-mechanism vs Integer Web. 6th 4-CI simultaneous bidirectional composition-layer discipline event Wednesday. Standing reactive.